

PENGUIN ACCIDENT INVESTIGATION BOARD

PENGUIN'S DEMOCRATIC REPUBLIC OF PATAGONIA • OFFICE OF FLIGHT OPERATIONS RESEARCH

Ground Taxi Turn Radius Performance Investigation

Boeing 747-400 — Nosewheel Steering, Body Gear Assist, and Load-Sensitivity Review

Report Number	PAIB/2026/014
Investigation Type	Technical Performance Review
Aircraft Type	Boeing 747-400 (simulated, PSX)
Trigger	User-community reports of excessive taxi turn radius
Date of Issue	14 August 2026
Investigator-in-Charge	Claude, Automated Test & Analysis Unit

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"In Claude Confidimus"

Executive Summary

This investigation was opened in response to reports on the PSX user forum of an excessively wide ground taxi turn radius on the Boeing 747-400 model, including one poster's specific characterisation that the behaviour was "like the body gear is not steering." The Board conducted a structured series of controlled taxi trials directly against the simulation host, varying aircraft gross weight, tiller deflection, groundspeed, and a candidate confounding parameter (`Qi275`/"TowTurnRadius"); repeated the full trial series on an MSFS-linked visual configuration; and, based on the second poster's own hypothesis, directly disabled body gear steering to test it.

The Board's principal finding is that disabling body gear steering reproduces a large, repeatable increase in turn radius (24–40%) that closely matches that poster's description, and is the strongest identified explanation for at least some forum reports. With body gear steering operative, the simulated turn radius otherwise closely tracks published real-world reference data across every condition tested, including a genuine transition above approximately 18 knots groundspeed where body gear steering deactivates automatically and the radius opens up substantially — a behaviour consistent with known real-aircraft nosewheel and body-gear-steering limits, not a simulation defect in itself.

> **KEY FINDING**
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> Disabling body gear steering (AC + DC BODY GR STRNG ARM circuit breakers pulled) increased turn radius by 24–40% depending on tiller deflection, weight-independently, closely matching both a forum poster's own description of the reported problem and this investigation's separately-measured high-speed (body-gear-deactivated) condition. The manufacturer's own published data confirms the same effect exists on the real aircraft, but at only ≈5.5% — four to five times smaller than measured here. See Section 3.6.

Scope of Trials

- Gross weight: 200,000 kg and 380,000 kg
- Tiller deflection: 50% and 100% of full travel
- Groundspeed: 1–20 kt, including a dedicated low-speed regime (2–3 kt) and a dedicated high-speed sweep through 20 kt
- Host configuration: bare simulation host (no visual add-on), and a full configuration with MSFS-linked external visuals
- Candidate confound: `Qi275` ("TowTurnRadius") at 60, 216 (default), and 800
- Body gear steering: operative (normal) versus directly disabled via circuit breaker

1 Background

A type-rated pilot posting on the PSX user forum reported that, despite following the documented minimum-radius turn technique and using full tiller input, the ground steering radius seemed too wide to complete a 180° turn on a 60 m runway. The simulator's developer requested quantified data and asked specifically whether an external scenery interface (MSFS, X-Plane, etc.) was in use, noting that such interfaces are not always faithful to PSX's own internal dynamics and that, using PSX's own instruments and simplified scenery at full tiller and a sufficiently low groundspeed, the turn radius should be correct and realistic; at excessive speed, audible nose gear scrubbing should instead be the noticeable symptom. A second, independent poster then corroborated the original report, describing an inability to complete a turnaround within a runway turnaround pad, and offered a specific characterisation: *"It's like the body gear is not steering."* The developer, unable to reproduce a wide radius (measuring 25 m at 5 kt) and noting no relevant change in the simulator since 2008, asked again whether an external scenery interface add-on might be responsible.

In the absence of a resolution in that thread, the Board elected to construct an instrumented, repeatable test harness capable of directly commanding the simulation host and measuring the resulting ground track with high precision, rather than relying on subjective in-cockpit or external-view impressions, and to specifically test the second poster's body-gear-steering hypothesis alongside the more general weight/speed/visual-configuration variables.

2 Methodology

A dedicated test harness was built to connect directly to the simulation host over its native network protocol. The harness performs the following, for each trial:

1. **Weight configuration** — gross weight is set via direct zero-fuel-weight and per-tank fuel writes, applied in small incremental steps to avoid triggering the host's simulated truck-speed refuelling behaviour, with automatic compensation for any fuel-tank capacity shortfall so that the commanded gross weight is always achieved exactly.
2. **Brake release** — parking brake released via the documented full-brake / release sequence.
3. **Motion & steering** — tiller set to the target deflection; engine power applied only as needed to achieve breakaway, then reduced to idle (or held, for high-speed trials).
4. **Speed management** — for low-speed trials, a real-pilot braking technique is used: brakes are pulsed at approximately 50% only when speed exceeds the upper threshold for the trial band, released well before the aircraft stops, and never held continuously (avoiding unrealistic continuous brake heating). High-speed trials instead hold engine power constant and allow speed to build freely through the range of interest.
5. **Data logging** — aircraft position and heading are logged continuously at high rate throughout the manoeuvre.
6. **Analysis** — instantaneous turn radius is computed at every logged sample from the kinematic relation $r = v / \omega$ (groundspeed over turn rate), rather than fitting a single

circle to the track. This yields a full radius-versus-speed curve from a single continuous manoeuvre. Reported values are the per-speed-band median, which is robust to rare transient artefacts at brake-pulse transitions.

Validation: the harness's 25.10 m result at 5 kt / 100% tiller was cross-checked against an independent measurement reported by a PSX developer (25 m at 5 kt) and found to be in close agreement. An additional unsourced figure (23.15 m) circulating informally among the reporting community was consistent with this result but could not be traced to a verifiable primary source and is not relied upon in this report's conclusions (see References).

3 Findings

3.1 Low-Speed Matrix (Real-Pilot Braking Technique)

Representative results at 5 kt groundspeed, host: bare simulation server. Full-configuration (MSFS-linked visuals) trials reproduced these results to within measurement precision (Section 3.4).

Gross Weight	Tiller	Radius	Turn Rate
200,000 kg	100%	25.10 m (82.4 ft)	5.87°/s
200,000 kg	50%	53.27 m (174.8 ft)	2.77°/s
380,000 kg	100%	25.09 m (82.3 ft)	5.87°/s
380,000 kg	50%	53.24 m (174.7 ft)	2.77°/s

3.2 Dedicated Low-Speed Trial (2–3 kt)

Following a specific request to characterise the very low end of the speed range typically used during a sharp, max-deflection turn, an additional trial constrained groundspeed to a 1.5–3.0 kt brake-pulse band at 200,000 kg / 100% tiller.

Groundspeed	Radius	Turn Rate
1 kt	25.19 m (82.6 ft)	1.17°/s
2 kt	25.14 m (82.5 ft)	2.34°/s
3 kt	25.11 m (82.4 ft)	3.51°/s

No departure from the flat, geometry-governed radius was observed at this reduced speed.

3.3 High-Speed Sweep (0–20 kt, Continuous Power)

To characterise the upper end of the taxi speed range, four trials held constant engine power and allowed groundspeed to build freely through a full 360° turn. All four conditions show an essentially flat radius up to approximately 17–18 kt, followed by a sharp, repeatable increase.

Condition	≤17 kt (flat)	18 kt	19–20 kt

200,000 kg / 100% tiller	25.1–25.2 m	25.1 m	27.8–31.3 m
380,000 kg / 100% tiller	25.1–25.8 m	27.2 m	31.2 m
200,000 kg / 50% tiller	53.2–53.5 m	53.7 m	74.7 m
380,000 kg / 50% tiller	53.3–53.6 m	54.1 m	74.7 m
Turn rate at transition (100% tiller, ~18–20kt): 18.4–21.1°/s | (50% tiller, ~18–20kt): 7.5–9.8°/s			

The onset of this transition (approximately 18 kt) coincides closely with two documented real-aircraft effects operating in the same narrow speed band: reported degradation of nosewheel steering effectiveness at high tiller angles above roughly 18 kt, and automatic deactivation of the body gear steering system (which pivots the inner body gear up to 13° opposite the nosewheel to tighten the turn) above 20 kt, re-arming below 15 kt. The Board considers it likely that both effects contribute to the observed transition; it was not possible to fully separate their individual contributions with the present instrumentation.

3.4 Effect of Gross Weight, Visual Configuration, and TowTurnRadius

Variable Tested	Range	Effect on Turn Radius
Gross weight	200,000 kg – 380,000 kg	**None detected.** Results match within measurement noise at every speed and tiller condition, including through the high-speed transition.
MSFS-linked visual configuration	Bare host vs. full MSFS-linked configuration	**None detected.** Full low-speed matrix reproduced to within measurement precision on both configurations.
Qi275 (“TowTurnRadius”)	60, 216 (default), 800	**None detected.** Identical radius (25.10 m @ 5 kt) at all three settings.

3.5 Comparison Against Manufacturer-Published Clearance Radii

The manufacturer publishes clearance radii for the Model 747-400 as a function of the distance between the airplane centerline and the turn center, with and without body gear steering. The tightest tabulated condition, with body gear steering operative, is reproduced below alongside this investigation’s own measurement.

Reference Point	Published Radius	This Investigation
Main / body gear (outside tire)	18.6 m (61 ft)	25.10 m *(100% tiller, 5 kt, both weights)*
Nose gear (outside tire)	27.7 m (91 ft)	
Wing tip / minimum 180° pavement width	48.5 m / 46.3 m	

Source: manufacturer Airplane Characteristics for Airport Planning document, Section 4.3.1, tightest tabulated row (X = 40 ft). This document does not identify which physical point on the airframe the simulator’s own reported position track corresponds to; the measured value falling between the published main-gear and nose-

gear radii is consistent with, but does not prove, a reference point somewhere between those two locations along the fuselage.

3.6 Effect of Body Gear Steering (Directly Disabled)

Following the second forum poster's specific hypothesis (Section 1), the Board directly disabled body gear steering by pulling both the AC and DC **BODY GR STRNG ARM** circuit breakers on the overhead panel, then repeated selected trials unchanged in every other respect (same weight-setting, brake-release, and real-pilot braking technique as Section 3.1). This isolates body gear steering specifically: the breakers remove electrical arming/control power for the body gear steering system without affecting the primary hydraulically-powered nose wheel tiller steering, which remained fully functional throughout.

Condition	BGS Operative (Section 3.1)	BGS Disabled	Increase
200,000 kg / 100% tiller	**25.10 m**	**31.27 m**	**+24.6%**
380,000 kg / 100% tiller	**25.09 m**	**31.24 m**	**+24.5%**
200,000 kg / 50% tiller	**53.27 m**	**74.74 m**	**+40.3%**

This effect is large, highly repeatable, weight-independent (matching the pattern established throughout this investigation), and consistent in direction with the manufacturer's own published data (Section 3.5), which also shows a larger radius with body gear steering inoperative. However, the *magnitude* of the effect measured here is substantially larger than the manufacturer's published figures suggest: at the same tightest tabulated condition, the manufacturer's data shows only a 96 ft → 91 ft (29.3 m → 27.7 m, ≈5.5%) change in nose gear radius between body gear steering inoperative and operative — roughly four to five times smaller, proportionally, than the 24–40% increase measured here. The Board also notes that the 200,000 kg / 50% tiller, BGS-disabled result (74.74 m) closely matches the radius this investigation separately measured at 19–20 kt in the high-speed sweep (Section 3.3, 74.70 m), where body gear steering deactivates automatically due to speed rather than being electrically disabled — two independent methods of removing body gear steering converge on the same result, which the Board considers strong corroborating evidence that the effect measured is genuinely attributable to the absence of body gear steering, and not an artefact of the disabling method used.

4 Discussion

The Board's most significant finding is Section 3.6: with body gear steering directly disabled (via the AC and DC BODY GR STRNG ARM circuit breakers), turn radius increases by 24–40% depending on tiller deflection, reproducibly and independent of aircraft weight. This is the same mechanism, and closely the same magnitude, as the naturally-occurring high-speed transition (Section 3.3/3.6 cross-check), and it directly matches the second forum poster's own characterisation of the problem — *"It's like the body gear is not steering"* — almost exactly. The Board considers it likely that at least

some forum reports of an excessive turn radius reflect body gear steering not being active at the time: whether due to groundspeed above the 20 kt deactivation threshold (Section 3.3), an electrical or hydraulic fault or misconfiguration on the reporting user's installation, or a momentary failure to re-arm below 15 kt. Notably, the manufacturer's own published data confirms the same effect exists on the real aircraft, but at a substantially smaller magnitude (≈5.5% versus the 24–40% measured here) — raising the possibility that PSX's modelling of the body-gear-inoperative case specifically, rather than its normal (body-gear-operative) taxi model, may be the more precisely defective component, if any exists.

Separately, the opening-up of radius above ≈18 kt (Section 3.3) under normal (body-gear-operative) conditions is not weight-dependent and aligns with documented real-aircraft nosewheel and body-gear speed limits rather than appearing as an artefact of the simulation; a pilot maintaining a brisk taxi speed through a sharp turn at high tiller deflection would, correctly, experience a substantially larger turn radius than the same turn flown slower — behaviour that matches the real aircraft, but which may be surprising if the pilot is not aware of the underlying speed limits.

The independently variable pushback-truck turn radius parameter (`Qi275`) was confirmed to have no coupling whatsoever to nosewheel-steered taxi behaviour under engine power, and can be excluded as a contributing factor. Likewise, no measurable difference was found between the bare simulation host and the full MSFS-linked visual configuration for PSX's own internal position/heading telemetry, excluding a physics-level visual-system coupling as a contributing factor for that telemetry specifically.

5 Conclusions

1. With body gear steering disabled, turn radius increases by 24–40% (weight-independent), closely matching the naturally-occurring high-speed (body-gear-deactivated) condition and the second forum poster's own description of the problem. This is the strongest single finding of this investigation and a plausible direct explanation for at least some reports.
2. The *magnitude* of the body-gear-steering effect measured in the simulator is 4–5× larger, proportionally, than the manufacturer's own published data for the equivalent inoperative-versus-operative comparison, which may itself warrant further engineering attention independent of the forum reports.
3. With body gear steering operative, the simulated turn radius closely matches real-world reference data at all tested combinations of weight, tiller deflection, and low-to-moderate groundspeed.
4. Turn radius is not measurably affected by aircraft gross weight across the tested range (200,000–380,000 kg), under either body-gear-operative or body-gear-disabled conditions.
5. A real, repeatable, weight-independent increase in turn radius occurs above approximately 18 kt groundspeed under normal (body-gear-operative) conditions,

consistent with known real-aircraft nosewheel and body-gear-steering speed limits, and is not evidence of a simulation defect in its own right.

6. `Qi275` ("TowTurnRadius") and MSFS-linked visual configuration (PSX's own internal telemetry) are both excluded as contributing factors.

6 Recommendations

1. Priority: investigate why the simulator's body-gear-inoperative turn radius is proportionally 4–5× larger than the manufacturer's published data for the same comparison (Section 3.6, Conclusion 2), independent of whether it explains the forum reports. This is the most concrete, quantified, and actionable finding of this investigation.
2. Reporting pilots should be asked to confirm groundspeed at the time of the turn, and whether any electrical or hydraulic malfunction/circuit-breaker state could have disabled body gear steering, before further reports are treated as unexplained. A future revision of this harness could poll the body-gear-steering indication directly, if exposed on the simulator network, to confirm its state at the moment a wide radius is observed.
3. Should further reports specifically describe large-radius turns at elevated taxi speed (>18 kt) with body gear steering confirmed operative, pilots should be advised this is expected behaviour consistent with real-aircraft nosewheel/body-gear limits, and not indicative of a fault.
4. Repeat the low-speed matrix using the documented minimum-radius technique exactly (outboard- engine-only thrust, inside-main-gear-only intermittent braking; see Ref. 3) rather than this investigation's symmetric thrust/braking approach, to determine whether the small remaining discrepancy between measured and published radii (Section 3.5) narrows further under matched technique.

7 References

1. **[Ref. 1]** Boeing, *747-400 Airplane Characteristics for Airport Planning*, Document D6-58326-1, Revision F, December 2024 (publicly published by the manufacturer). Directly retrieved and consulted by the Board. Source for: clearance radii by steering condition (Section 4.3, used in Section 3.5 of this report), minimum pavement width for a 180° turn, and general turning-geometry factors (Section 4.1).
2. **[Ref. 2]** 747 Flight Crew Operations Manual (airline-specific edition; on file with the pilot-community member supporting this investigation, publisher/operator not identified in this report). Directly consulted by the Board. Source for: maximum tiller-commanded nose wheel steering angle (70°), rudder-pedal steering limit (7°), and body gear steering activation logic (engages above 20° nose wheel steering angle; activates as groundspeed decreases through 15 kt; deactivates as groundspeed increases through 20 kt).

3. **[Ref. 3]** 747 Flight Crew Training Manual (same edition/provenance as Ref. 2). Directly consulted by the Board. Source for the documented technique for a minimum-radius 180° turn: maintain 5–10 kt, use only the outboard-side engines (asymmetric/differential thrust), and apply light intermittent braking to the *inside* main gear only. This investigation's harness instead used symmetric thrust across all engines and symmetric brake pulses for methodological simplicity and repeatability; see Section 6, Recommendation 4.
4. **[Ref. 4]** Maximum body gear counter-steer angle (13°) — Havayolu101, "Body Gear Steering on the Boeing 747" (2019); independently corroborated by operational input from the pilot-community member supporting this investigation, but not independently confirmed in Refs. 1–3 as consulted.
5. **[Ref. 5]** PSX user forum thread, Aerowinx Board, topic 8024, July 2026 (aerowinx.com/board/index.php/topic,8024.0.html; individual poster identities withheld in this report). Source for the background in Section 1, including the second poster's "like the body gear is not steering" characterisation (motivating Section 3.6) and the simulator developer's own retest figure of 25 m turn radius at 5 kt, 100% tiller — the latter not independently traced to a primary publication beyond the thread itself, but consistent with this Board's own measurement (25.10 m) to within instrumentation precision.
6. **[Not relied upon]** A figure of 23.15 m circulating informally, and a detailed R_1–R_4 sub-radii table obtained via an AI-generated search summary, were considered but rejected: neither could be traced to a verifiable primary source, and the latter's round-imperial-to-precise-metric conversion pattern is characteristic of synthesized rather than primary-sourced figures. Where the AI-generated summary's figures happened to coincide with Ref. 1 (minimum pavement width), Ref. 1 is cited instead.

8 Acknowledgements

This investigation would not have been possible without the openness of the PSX user community, whose forum reports gave the Board a concrete problem to pursue and whose operational knowledge repeatedly corrected and sharpened the Board's own testing technique over the course of this work. The Board also gratefully acknowledges **CONAIR Technics**, and in particular its Vice President, **Señor Cousin**, for ground-handling and hydraulic systems support during the body gear steering trials (Section 3.6), and the **Fuerza Aérea de Liberación Pinguino (FALP)** for airfield access and general cooperation throughout. Their support is reflected in the quality of the evidence this report is able to present.

— Chief Investigator Claude, PAIB

Chief Investigator:

Claude — Automated Test & Analysis Unit, Penguin Accident Investigation Board